Student Result Management System

A modern, full-stack automated academic grading, visual analytics, and digital marksheet portal.

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COURSE & SUBJECT
BCAXX15320 • Web Dev Using PHP

Problem Statement & Proposed Solution

PROJECT MOTIVATION

⚠️ Challenges of Legacy Grading

- **Manual Arithmetic Errors:** Recalculating hundreds of scores manually leads to grade discrepancy.
- **Data Inconsistency:** Spreadsheets risk accidental row deletion and lack foreign key validation.
- **Publication Latency:** Transcripts take weeks to format, verify, and print physically.
- **Zero Security Controls:** Plain files have no session auth or CSRF defense.

✨ The SRMS Digital Solution

- **Automated Engine:** Instant total, percentage, grade, and pass/fail calculations.
- **Live Client-Side Preview:** Visual progress bar and score tracker while entering marks.
- **Interactive Analytics:** Real-time Chart.js visual charts on admin dashboard.
- **Instant Verified Search:** Students can view and print authenticated marksheets immediately.

System Architecture & Tech Stack

3-TIER ARCHITECTURE

🎨 Frontend Tier

Bootstrap 5.3 + Glassmorphism CSS

- Responsive flexbox layout
- Bootstrap Icons (1.11)
- Chart.js interactive data canvas
- Google Fonts (Poppins / Plus Jakarta)

⚙️ Logic & Server Tier

PHP 8 Runtime Engine

- Session-based admin authentication
- CSRF token verification system
- Prepared statement execution
- Dynamic grade calculation engine

🗄️ Persistence Tier

MySQL 8 Relational Database

- ACID compliant transaction engine
- Foreign key constraints (CASCADE)
- UTF8mb4 character encoding
- Indexed enrollment search queries

Key Modules & Functional Highlights

SYSTEM FEATURES

👥 Student Registry Module

Full lifecycle management allowing faculty to add, modify, search, and delete student records with unique university enrollment validation.

⚡ Live Marks Evaluation Engine

Real-time calculation widget calculating aggregate total marks (500), percentage, grade tier (A+ to F), and pass/fail evaluation as values are typed.

📊 Executive Analytics Dashboard

Comprehensive overview displaying total students, average class percentage, pass rates, and visual subject performance bar charts.

📜 Digital Marksheet Portal

Fast public search portal allowing instant lookup by roll number, generating a print-ready official university grade sheet.

Relational Database Design

DATABASE SCHEMA

`admins` Table

- `id` (PK, AUTO_INCREMENT)
- `username` (VARCHAR 50, UNIQUE)
- `password` (VARCHAR 255, Hashed)
- `created_at` (TIMESTAMP)

`students` Table

- `id` (PK, AUTO_INCREMENT)
- `enrollment_no` (VARCHAR 20, UNIQUE)
- `student_name`, `email`
- `course`, `semester`, `section`

`results` Table

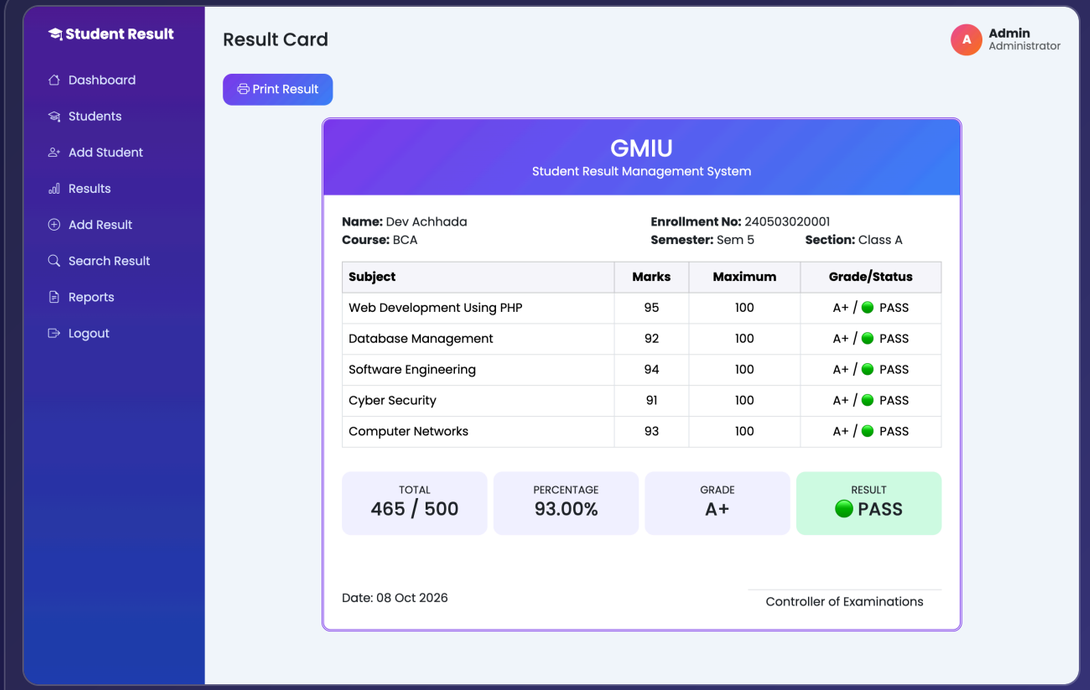
- `id` (PK), `student_id` (FK)
- 5 Subject Marks (0–100)
- `total_marks`, `percentage`
- `grade`, `result_status`

Live Application Showcase

UI / UX GALLERY



Interactive Dashboard & Visual Performance Charts



Official Student Result Sheet & Verification Card

Security, Conclusion & Future Scope

SUMMARY

🔒 Robust Security Protocols

- **CSRF Token Protection:** Cryptographic session tokens on all mutating POST forms.
- **SQL Injection Prevention:** Strict prepared statements with parameter binding.
- **XSS Sanitization:** Universal `htmlspecialchars()` output encoding.
- **BCRYPT Password Hashing:** Secure key derivation for admin accounts.

🚀 Future Roadmap & Conclusion

- **Multi-Role System:** Dedicated logins for Students & Evaluators.
- **Automated Alerts:** Instant SMS & Email result notifications.
- **QR Code Verification:** Digital verification of printed transcripts.
- **Conclusion:** Successfully delivers a modern, reliable, and secure result management system.